EE1101: Circuits and Network Analysis

Assignment - 12

Handed out: 02 - Nov - 2024 **Due : 12 - Nov - 2024 (before 5 PM)**

Instructions :

1. Please upload your assignment solutions to the course page on the Canvas platform. Only solutions submitted through canvas will be reviewed. For specific guidelines, refer to the instructions provided on the course page.
2. It is suggested that you attempt all the problems. However, it is sufficient to submit solutions for problems that total 10 points.
3. Submissions received after the deadline will attract negative marking. Ensure that your submissions are named in the following format: RollNo-Assignment-13.pdf.

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1. (5 points) Consider the circuit shown in Fig. 1.

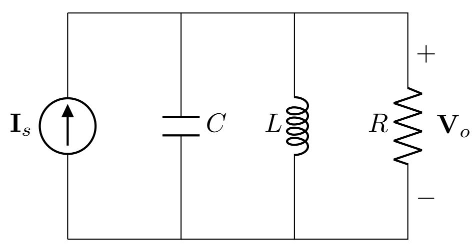

Fig. 1: Circuit for question 1

   (a) (2 points) Compute the $s-$ domain transfer function $\mathbf{H}(\mathbf{s}) = \frac{\mathbf{V}_o}{\mathbf{I}_s}$.

   (b) (2 points) Plot the approximate (asymptotic bode plot) magnitude response of the transfer function for $R = 1\Omega$, $L = 1$H and $C = 1\mu$F.

   (c) (1 point) Compute the corner frequency and bandwidth of the transfer function for $R = 1\Omega$, $L = 1$H and $C = 1\mu$F.

2. (5 points) Consider the circuit shown in Fig. 2.

   (a) (2 points) Compute the $s-$ domain transfer function $\mathbf{H}(\mathbf{s}) = \frac{\mathbf{V}_o}{\mathbf{V}_i}$.

   (b) (2 points) Plot the approximate (asymptotic bode plot) magnitude response of the transfer function for $R = 1\Omega$, $L = 1$H and $C = 1\mu$F.

   (c) (1 point) Compute the corner frequency and bandwidth of the transfer function for $R = 1\Omega$, $L = 1$H and $C = 1\mu$F.

3. (5 points) Consider the circuit shown in Fig. 3.

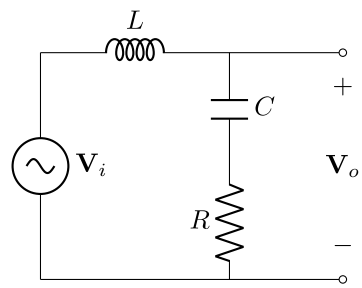

Fig. 2: Circuit for question 2

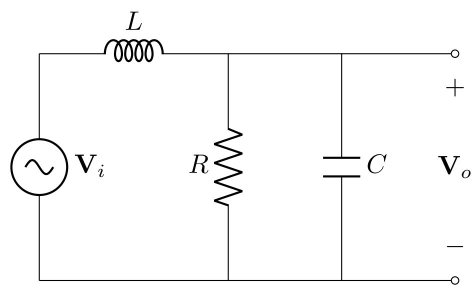

Fig. 3: Circuit for question 3

(a) (2 points) Compute the $s-$ domain transfer function $\mathbf{H(s)} = \frac{\mathbf{V}_o}{\mathbf{V}_i}$.

(b) (2 points) Plot the approximate (asymptotic bode plot) magnitude response of the transfer function for $R = 1\Omega$, $L = 1\text{H}$ and $C = 1\mu\text{F}$.

(c) (1 point) Compute the corner frequency and bandwidth of the transfer function for $R = 1\Omega$, $L = 1\text{H}$ and $C = 1\mu\text{F}$.

4. (5 points) Consider the circuit shown in Fig. 4.

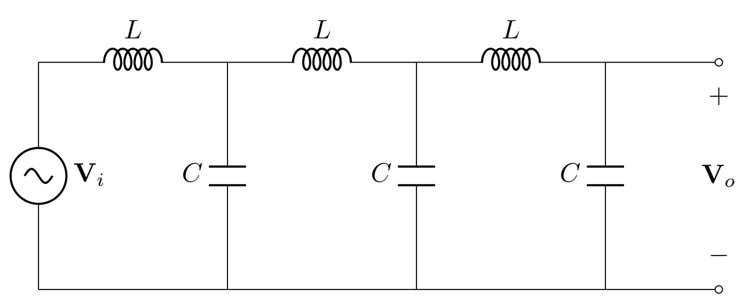

Fig. 4: Circuit for question 4

(a) (2 points) Compute the $s-$ domain transfer function $\mathbf{H(s)} = \frac{\mathbf{V}_o}{\mathbf{V}_i}$.

(b) (2 points) Plot the approximate (asymptotic bode plot) magnitude response of the transfer function for $L = 1\text{H}$ and $C = 1\mu\text{F}$.

(c) (1 point) Compute the corner frequency and bandwidth of the transfer function for $L = 1\text{H}$ and $C = 1\mu\text{F}$.